

Discrete Mathematics Assignment 2

Due Date: March 25, 2026

Write clear and concise handwritten solutions. You must also upload a soft copy of it on the moodle page. It is fine to discuss with others, but your solutions must be in your own words that you have fully understood.

1. This exercise is about counting rooted labeled trees on n vertices. The goal is to show that this number t_n is n^{n-1} using exponential generating functions as outlined in class. Let $T(x)$ denote the exponential generating function for the sequence $\{t_n\}_{n \geq 1}$. Show the equation $T(x) = xe^{T(x)}$. Using formal derivatives find a way of solving for $T(x)$ from this equation and hence derive that $t_n = n^{n-1}$.
2. A forest is a graph without any cycles. Find the number of labeled forests on n vertices with exactly k *rooted trees* by first obtaining its exponential generating function.
3. This exercise is about counting the number of labeled forests on n vertices (but now the forest consists of *unrooted* trees). Let $F(n)$ denote the number of labeled forests on vertex set $[n]$. Show the following recurrence relation holds:

$$F(n) = \sum_{k=1}^n \binom{n-1}{k-1} k^{k-2} F(n-k).$$

4. Recall the inclusion-exclusion formula

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{|I|=1}^n (-1)^{|I|-1} |A_I|,$$

discussed in class. Show for every odd $k \leq n$ that $|\bigcup_{i=1}^n A_i| \leq \sum_{|I|=1}^k (-1)^{|I|-1} |A_I|$ and for every even k that $|\bigcup_{i=1}^n A_i| \geq \sum_{|I|=1}^k (-1)^{|I|-1} |A_I|$.

5. Are the infinite products $\prod_{n \geq 1} (1 + x^{2^n})$ and $\prod_{n \geq 1} (1 - x^{2^n})$ well defined as generating functions? If so, prove it and find a combinatorial meaning to the coefficients of the resulting generating functions.

6. This exercise is on applying Möbius inversion for the divisibility poset. The n^{th} cyclotomic polynomial is defined as $\Phi_n(x) = \prod_{\omega} (x - \omega)$ where the product is over all *primitive* n^{th} roots of unity. Show using Möbius inversion that $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(d,n)}$.
7. Show that $\sum_{d^2|n} \mu(1, d)$ is 1 if n is square-free and zero otherwise. Using this along with Möbius inversion for the divisibility poset derive the following formula for counting the number $S(x)$ of square free positive integers bounded by x :

$$S(x) = \sum_{d=1}^{\lfloor \sqrt{x} \rfloor} \mu(1, d) \lfloor \frac{x}{d^2} \rfloor.$$

8. Consider permutations $p_1, p_2, \dots, p_n$ on $\{1, 2, \dots, n\}$ that avoid all triples $i < j < k$ such that $p_i < p_j < p_k$. Show that the number of such permutations is the n^{th} Catalan number C_n . You can either give a bijective proof or directly solve via recurrence.